



tom-analytics

One-Line Autonomous Data Analytics & EDA Library

A highly optimized, zero-configuration Python library that delivers end-to-end automated exploratory data analysis, robust statistical diagnostics, and premium reports from a single line of code.

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Version:

v0.1.0 (Stable)

Platform:

Windows / macOS / Linux

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1. Introduction & Overview

In modern data workflows, exploratory data analysis (EDA) is an essential first step. However, writing repetitive parsing, cleaning, plotting, and statistical analysis code is time-consuming. The tom-analytics library solves this by

Operational Flow (Under the Hood)

1. File Ingestion: Automatic delimiter, format, and encoding detection via `om.file()`.
2. Cleaning: Type coercion, median/mode imputation, drop columns with >50% nulls, drop duplicates.
3. Statistical Engine: Executes normality (Shapiro-Wilk), Chi-Square association, and ANOVA tests.
4. Visual Suite: Generates 300 DPI high-res Seaborn distributions, Q-Q, violin plots, and outliers.
5. Glassmorphism UI: Assembles a single, shareable HTML report with base64 embedded assets.

Key Features & Advantages

- Universal Loaders:** Delimiter-independent parser supporting CSV, TSV, JSON, Excel, and Parquet out-of-the-box.
- Windows Resiliency:** Automatically configures `sys.stdout` to UTF-8 on load to eliminate Windows CP1252 character-encoding console crashes completely.
- Advanced Statistics:** Distinguishes parametric vs non-parametric metrics and chooses statistical tests dynamically based on normality tests.
- Interactive Visuals:** Produces both static Seaborn visualizations and interactive HTML-embedded Plotly dashboards allowing zoom, hover, and panning.

2. Installation & Quickstart

```
pip install tom-analytics
```

Installation from PyPI

Local Development / Editable Installation

```
git clone https://github.com/OmkarSugave/tom.git
cd tom && pip install -e .
```

Quickstart Pipeline

```
# Import and load dataset
import tom as om
om.file("dataset.csv")
# Generate report.html and terminal diagnostics
om.describe()
```

3. Deep API Reference

om.file(path=None)

Loads any tabular file. Automatically infers formats, delimiters, column headers, and schemas. If no path is provided, launches an interactive terminal dialogue. Features intelligent edit-distance spellcheck/suggestions for path mistakes.

om.describe()

The primary entry point. Executes data cleaning, computes ANOVA/Chi-Square statistical tests, generates 300 DPI high-resolution figures, builds an interactive Plotly dashboard, and compiles a premium backdrop-blur glassmorphic HTML report.

om.clean_report(df=None)

Prints a high-fidelity summary detailing pre-processed columns. Displays original vs. inferred types, duplicate rows dropped, missing value counts, outlier percentages, and lists auto-imputed metrics.

om.correlation(df=None)

Calculates the Pearson correlation matrix for numeric columns, displays a formatted table in the console, and exports a high-resolution heatmap at `./tom_report/charts/single_correlation_heatmap.png`.

om.outliers(df=None)

Computes a robust outlier diagnostic using the Interquartile Range (IQR) method. Displays upper/lower limits, min/max outliers, and lists total counts and percentages for each numerical variable.

om.suggest(df=None)

Autonomous machine learning workflow advisor. Analyzes column categories to automatically target suitable algorithms. Determines whether a regression or classification pipeline is required, suggests candidate ML models (e.g. Random Forests, XGBoost, Ridge, Logistic Regression), and lists custom pre-processing rules.

om.compare(df2, df1=None)

Compares two datasets side-by-side (e.g. training set vs test set). Generates a metrics overview detailing shapes, duplicates, missing cells, and displays specific overlapping columns or unique fields.

om.export(format_type='pdf')

Converts the generated glassmorphic HTML report into a standard PDF page. Gracefully lists instructions if local packages like `pdftkit` or system `wkhtmltopdf` binaries are missing.

```
tom/                                # Project Root
|-- tom/                            # Core Package Modules
| |-- __init__.py                   # Package State & Standard UTF-8 Reconfig
| |-- loader.py                    # Format Independent Smart Loader Engine
| |-- cleaner.py                   # Median/Mode Imputers & Clean Logs
| |-- stats.py                     # Shapiro-Wilk & Parametric Hypothesis Suite
| |-- charts.py                    # High-DPI Seaborn Suite & Base64 Encoder
| |-- insights.py                  # NLP Multi-Collinearity & Skew Warnings
| |-- reporter.py                  # Glassmorphic HTML & Plotly Compiler
| |-- utils.py                     # Spellcheck Levenshtein Distance & Dirs
|-- pyproject.toml                  # Setuptools Package Metadata & Dependencies
+-- tom_report/                     # Outputs (Excluded in .gitignore)
    |-- report.html                 # Portable Backdrop-Blur HTML Report
    +-- charts/
        +-- plotly_dashboard.html # Full Pan & Zoom Plotly Analytics
```

License & Credits

The tom-analytics package is open-source software licensed under the MIT License. It relies heavily on powerful open-source libraries in the Python scientific ecosystem, including Pandas, Numpy, Scipy, Matplotlib, Seaborn, Rich, and Plotly. For additional support, bug reporting, or code contributions, please visit the repository at: <https://github.com/OmkarSugave/tom.git>

End of Official Documentation - Build tom-analytics